

MCGREGOR, MN, USA

WMO: 720258

Lat: 46.619N

Lon: 93.310W

Elev: 374

StdP: 96.91

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04997

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -27.4	(c) -25.0	(d) -31.2	(e) 0.2	(f) -26.6	(g) -27.9	(h) 0.3	(i) -23.6	(j) 9.0	(k) -9.7	(l) 8.3	(m) -9.1	(n) 1.5	(o) 310	(p) 0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.9	(c) 29.9	(d) 21.9	(e) 27.7	(f) 20.1	(g) 27.0	(h) 19.5	(i) 23.1	(j) 28.1	(k) 21.9	(l) 26.3	(m) 20.7	(n) 24.3	(o) 4.4	(p) 160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.4	16.8	26.0	20.2	15.6	24.7	19.1	14.5	23.5	70.5	27.8	65.9	27.0	61.5	24.9	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.4	(b) 7.6	(c) 7.0	(e) -31.9	(f) 32.3	(g) 3.1	(h) 1.5	(i) -34.2	(j) 33.4	(k) -36.0	(l) 34.3	(m) -37.7	(n) 35.2	(o) -40.0	(p) 36.3		
(a) 8.4	(b) 7.6	(c) 7.0	(e) -32.1	(f) 25.2	(g) 3.1	(h) 1.3	(i) -34.3	(j) 26.1	(k) -36.1	(l) 26.9	(m) -37.8	(n) 27.7	(o) -40.0	(p) 28.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.3	-10.9	-9.5	-1.7	5.3	11.7	16.9	19.9	18.1	14.3	7.1	-0.3	-8.1		
	DBStd	11.95	7.65	6.69	6.79	5.06	4.44	3.39	3.14	3.16	4.40	4.49	5.60	6.57		
	HDD10.0	2743	649	545	365	155	32	1	0	0	11	114	311	559		
	HDD18.3	4897	908	778	620	392	212	66	19	42	134	348	559	818		
	CDD10.0	1034	0	0	3	13	83	207	308	253	140	24	2	0		
	CDD18.3	146	0	0	0	0	6	22	69	36	13	1	0	0		
	CDH23.3	1511	0	0	0	2	102	256	681	344	117	9	0	0		
	CDH26.7	298	0	0	0	0	21	38	160	62	16	1	0	0		
(14) Wind	WSAvg	2.9	2.9	2.9	3.0	3.5	3.2	2.8	2.4	2.3	2.8	3.0	3.1	2.8		
(15) Precipitation	PrecAvg	732	19	17	31	56	79	111	118	87	81	82	33	21		
	PrecMax	893	50	40	74	174	145	183	201	196	160	201	84	37		
	PrecMin	489	4	3	8	17	37	33	60	34	24	14	4	6		
	PrecStd	104	12	9	17	33	31	38	45	43	36	45	22	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.8	7.7	19.9	22.5	29.9	30.0	32.3	31.0	28.7	24.9	17.2	6.0		
		MCWB	2.6	4.5	15.3	13.1	19.4	21.9	23.7	23.0	20.9	15.7	11.4	3.8		
	2%	DB	2.3	3.9	14.1	19.9	26.2	27.5	30.2	27.9	26.4	20.3	12.6	2.8		
		MCWB	1.0	1.8	9.3	11.4	16.5	19.4	22.3	20.7	19.4	14.3	8.6	1.4		
	5%	DB	1.0	2.0	10.1	17.3	22.9	26.2	27.9	27.0	24.8	17.3	10.0	1.3		
		MCWB	0.4	0.6	6.6	9.7	14.5	18.1	20.6	19.8	18.6	12.1	6.7	0.7		
	10%	DB	-0.1	0.0	7.1	14.8	21.0	24.9	27.2	25.2	22.3	14.8	7.4	0.2		
		MCWB	-0.6	-0.9	4.2	8.1	13.8	17.3	19.9	18.6	17.5	10.6	5.3	-0.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.9	5.1	15.3	14.7	21.7	23.0	25.2	23.6	22.7	17.6	12.4	4.6		
		MCDB	5.9	8.2	18.5	20.2	26.4	28.0	29.6	28.6	26.9	21.2	15.2	5.4		
	2%	WB	1.2	2.1	9.9	12.2	18.3	21.2	23.6	22.4	21.0	15.1	9.1	1.9		
		MCDB	2.1	3.6	13.8	17.6	23.3	25.5	28.7	26.9	24.0	19.7	12.3	3.0		
	5%	WB	0.1	0.3	6.6	10.2	16.5	19.7	22.3	21.2	19.6	12.9	7.1	0.8		
		MCDB	0.5	1.3	9.8	16.3	21.3	24.6	27.0	24.7	23.5	16.5	9.1	1.2		
	10%	WB	-1.0	-1.0	4.3	8.5	14.7	18.4	21.0	20.0	18.1	11.1	5.3	-0.1		
		MCDB	-0.6	-0.2	6.9	14.2	19.6	22.7	24.8	23.8	21.3	13.7	7.9	0.2		
(35) Mean Daily Temperature Range	MDBR		9.4	10.4	10.2	11.0	12.4	12.0	11.9	11.9	11.4	9.2	7.7	7.6		
	5% DB	MCDBR	8.6	10.2	12.9	16.3	16.8	14.4	13.1	13.5	14.0	14.0	11.9	7.5		
		MCWBR	7.4	8.1	8.3	9.3	8.6	7.5	6.5	7.0	7.9	8.2	7.8	6.6		
	5% WB	MCDBR	8.2	9.4	12.1	14.9	13.8	12.3	11.9	11.2	11.7	11.8	10.7	6.9		
		MCWBR	7.3	7.7	8.2	9.0	8.2	7.4	6.7	6.7	7.4	7.9	7.5	6.3		
(40) Clear-Sky Solar Irradiance	taub		0.248	0.262	0.305	0.359	0.380	0.378	0.379	0.386	0.358	0.309	0.285	0.260		
	taud		2.391	2.359	2.374	2.340	2.334	2.365	2.381	2.382	2.436	2.569	2.515	2.415		
	Ebn at Noon		868	924	923	897	886	887	880	859	856	861	817	808		
	Edh at Noon		69	88	102	117	122	119	115	111	96	72	62	61		
(44) All-Sky Solar Radiation	RadAvg		1.56	2.68	3.80	4.65	5.33	5.78	6.16	5.16	3.89	2.38	1.47	1.10		
	RadStd		0.17	0.23	0.36	0.45	0.40	0.40	0.25	0.40	0.29	0.32	0.15	0.10		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page